

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Generate Histogram of Images , apply Histogram Equalization and Histogram matching on it	
Experiment 8	Date: 20/08/2026	Enrolment No:92400133050

GitHub Link :

Aim

Generate Histogram of Images , apply Histogram Equalization and Histogram matching on it

Code

```
import cv2

import numpy as np

from matplotlib import pyplot as plt

# Load the image

image_path = '/content/jonatan-pie-h8nxGssjQXs-unsplash.jpg' # Replace with the path to your image

image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

# Calculate the histogram

histogram = cv2.calcHist([image], [0], None, [256], [0, 256])

# Plot the histogram

plt.figure(figsize=(8, 6))

plt.title('Histogram')

plt.xlabel('Pixel Value')

plt.ylabel('Frequency')

plt.plot(histogram)

plt.xlim([0, 256])

plt.grid(True)

plt.show()

# Perform histogram equalization
```

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```
equalized_image = cv2.equalizeHist(image)
```

```
# Display the original and equalized images
```

```
plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
```

```
plt.title('Original Image')
```

```
plt.imshow(image, cmap='gray')
```

```
plt.axis('off')
```

```
plt.subplot(1, 2, 2)
```

```
plt.title('Equalized Image')
```

```
plt.imshow(equalized_image, cmap='gray')
```

```
plt.axis('off')
```

```
# Calculate the histogram
```

```
histogram = cv2.calcHist([equalized_image], [0], None, [256], [0, 256])
```

```
# Plot the histogram
```

```
plt.figure(figsize=(8, 6))
```

```
plt.title('Histogram')
```

```
plt.xlabel('Pixel Value')
```

```
plt.ylabel('Frequency')
```

```
plt.plot(histogram)
```

```
plt.xlim([0, 256])
```

```
plt.grid(True)
```

```
plt.show()
```

```
plt.tight_layout()
```

```
plt.show()
```

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```

# Load the source and reference images

source_path = '/content/varun-singh-RmWkrjjz2J4-unsplash.jpg'
reference_path = '/content/jonatan-pie-h8nxGssjQXs-unsplash.jpg'

source_image = cv2.imread(source_path, cv2.IMREAD_GRAYSCALE)
reference_image = cv2.imread(reference_path, cv2.IMREAD_GRAYSCALE)

# Calculate histograms for the source and reference images

source_hist = cv2.calcHist([source_image], [0], None, [256], [0, 256])
reference_hist = cv2.calcHist([reference_image], [0], None, [256], [0, 256])

# Normalize histograms to have sum equal to 1

source_hist /= source_hist.sum()
reference_hist /= reference_hist.sum()

# Calculate cumulative distribution functions (CDF) for histograms

source_cdf = source_hist.cumsum()
reference_cdf = reference_hist.cumsum()

# Perform histogram matching by mapping source CDF to reference CDF

mapping = np.interp(source_cdf, reference_cdf, range(256))
matched_image = mapping[source_image]

# Convert to uint8 data type

matched_image = matched_image.astype(np.uint8)

# Display the images using Matplotlib

plt.figure(figsize=(12, 6))

```

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```
plt.subplot(131)
```

```
plt.title('Source Image')
```

```
plt.imshow(source_image, cmap='gray')
```

```
plt.axis('off')
```

```
plt.subplot(132)
```

```
plt.title('Reference Image')
```

```
plt.imshow(reference_image, cmap='gray')
```

```
plt.axis('off')
```

```
plt.subplot(133)
```

```
plt.title('Matched Image')
```

```
plt.imshow(matched_image, cmap='gray')
```

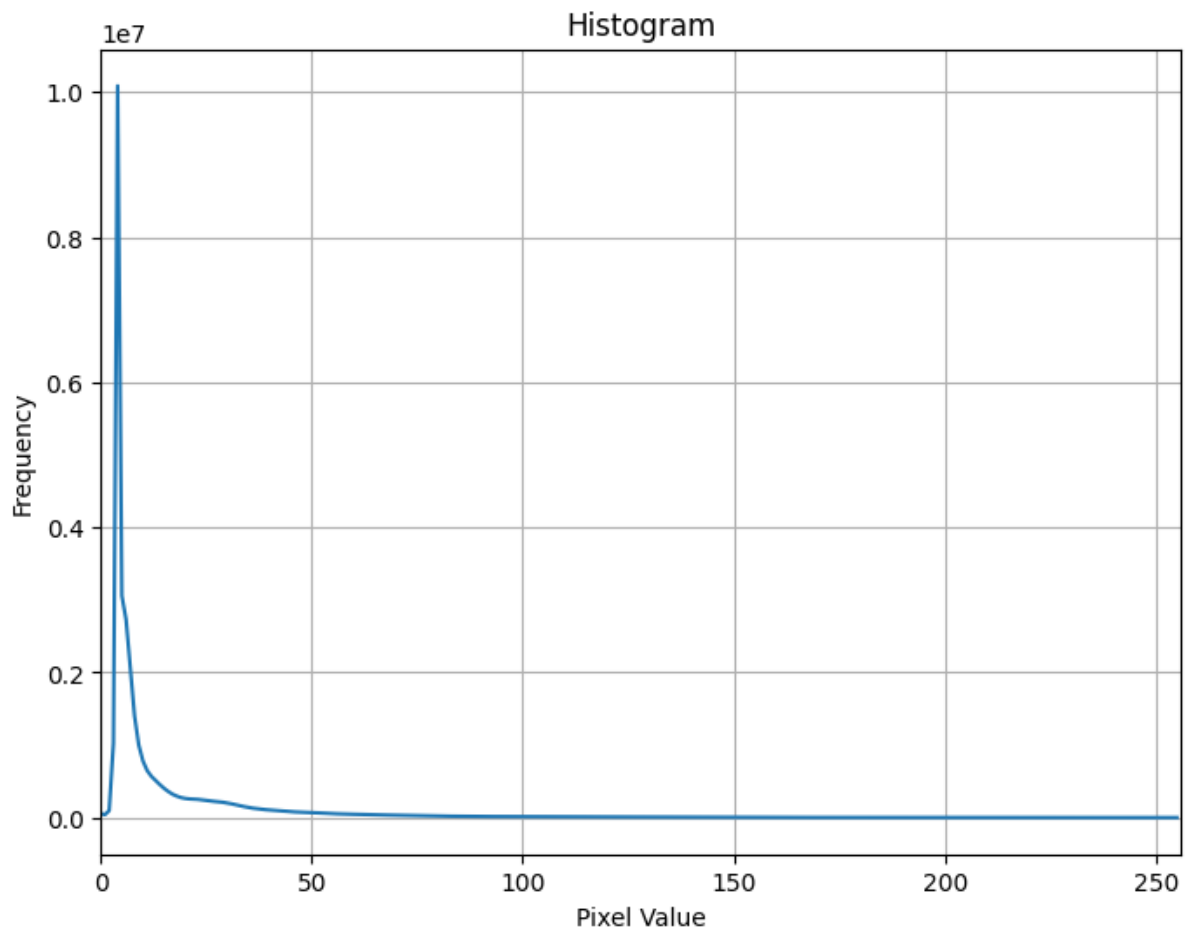
```
plt.axis('off')
```

```
plt.tight_layout()
```

```
plt.show()
```

Output

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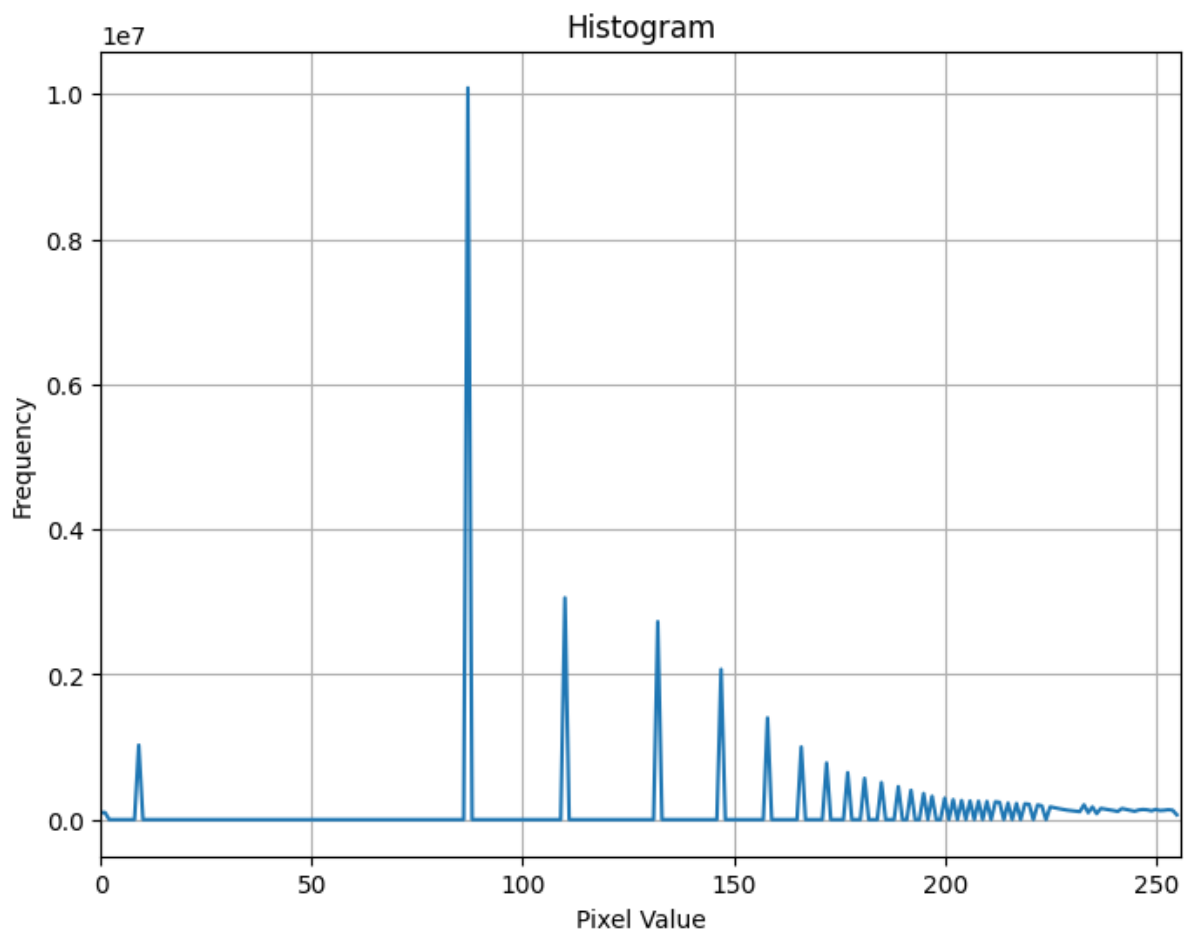
Original Image



Equalized Image



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Conclusion

This experiment helped us understand how histograms are used to analyze and improve grayscale images. Histogram equalization improved the image contrast by spreading the pixel intensity values, making important details more visible. Histogram matching was also performed using a reference image to adjust the intensity distribution of the source image. Overall, the experiment provided a practical understanding of histogram-based image enhancement and contrast improvement.